

HUMAN HEALTH RISK ASSESSMENT FROM THE CONTAMINANTS OF MAHANANDA RIVER WATER AND ITS ADJACENT GROUNDWATER

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Author's Name And affiliations: (9 font, First and Middle Name-abbreviation, Surname-Full, Use footnote for affiliations)

Abstract

This study aims to assess the physicochemical parameters of the surface water of the Mahananda River and the concentration of the nutrients of the Mahananda River water and the adjacent groundwater. Among physicochemical parameters, pH, Dissolved Oxygen (DO), and Total Dissolved Solids (TDS) were measured on 18 sampling sites of the river, while Sulfate (SO_4^{2-}), Orthophosphate (PO_4^{3-}), Arsenic (As), and Manganese (Mn) concentrations were measured on 17 equidistant sampling sites of the river and the adjacent groundwater. The mean pH value was 8.50 with a standard deviation (SD) of 0.30. DO and TDS were 9.56 mgL^{-1} (SD 1.67) and 127.62 mgL^{-1} (SD 24.76), respectively. The mean SO_4^{2-} concentration was found to be 5.38 mgL^{-1} (SD 12.00) for groundwater and 2.29 mgL^{-1} (SD 1.96) for surface water, while the mean PO_4^{3-} concentration was 1.17 mgL^{-1} (SD 0.78) and 0.63 mgL^{-1} (SD 0.15) for groundwater and surface water respectively. The contamination of As was found to be as high as $100 \text{ }\mu\text{gL}^{-1}$ in the groundwater while within $25 \text{ }\mu\text{gL}^{-1}$ in the surface water. Mn concentration was high at almost all the points, with a mean value of 1.64 mgL^{-1} (SD 0.90) and 0.54 mgL^{-1} (SD 0.23) for the groundwater and surface water, respectively. The carcinogenic risk from As and non-carcinogenic risk from both As and Mn was estimated for adults and children. The carcinogenic risks were found to be high for both surface water and groundwater, while non-carcinogenic risks were found high for surface water and very high for groundwater.

Abstract: Not more than 250 words, 8 font

Keywords: Arsenic, Carcinogenic Risk, Groundwater, Mahananda River, Manganese.

Keywords: 6-8 key words (8 font, Italic)

Introduction

Section heading: 10 font, bold

Mahananda is one of the major transboundary rivers between Bangladesh and India. It originates from the Himalayas. Sev... the Mal... read... Ban... point... barrage, where the river meets the Teesta irrigation canal of India and flows further downstream with the same canal name up to Nitaiganj. The mainstream of the river with the remaining streamflow flows as the border river for about 20 km. It then flows through India for 225 km and finally falls on the Ganges near the Farakka barrage. Prior to the falls on the... out 71 km from the district of Pabna. It is a tributary of the Ganges. The origin of the river is at Gomostapur, Punarbhaha River, and the other on the right bank at Chapainawabganj Sadar, Pagla River. According to the Bangladesh Water Development Board (BWDB) data, the river's minimum water level declined from 1977 to 2017. BWDB data also suggest that the adjacent groundwater level follows the same trend, and the southern part has a lower groundwater level than the northern part (RRI, 2019). The river contributes to the irrigation of above 10000 hectares of cropland. People near the river use its water for bathing,

washing, and household work. According to the Bangladesh Population and Housing Census 2011, about 8.3% of the district's population is engaged in agriculture, while 88.3% is engaged in non-agriculture, including water for... As is found to be much higher than the permissible limit of 10 mgL^{-1} in the groundwater. Inorganic arsenic is a term exposure, and hard cancer, like the bladder cancer, of As is a risk factor for diabetes, (Ravenscroft et al., 2005).

Methodology

Study Area

Sub Section Heading: 9 font, Italic

The whole reach of the Mahananda River in the Chapainawabganj district of Bangladesh....

Sampling Points

Physicochemical parameters were analyzed at 18 points of the Mahananda River..... e river sampling points. The surface water sampling points for physicochemical parameters are shown in Fig. 1 and Table 1, while nutrients

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Corresponding author's email only

and elemental analysis points are shown in **Fig. 2** and **Table 2**.

Health Risk Assessment

the ingestion route (I_o) was calculated by Eq. (1) (USEPA, 1989):

$$I_o = \frac{C_w \times IR \times EF \times ED}{BW \times AT} \quad \text{Eq. (1)}$$

where C_w is the metal concentration at the exposure site
dermal contact (I_d) was calculated by Eq. (2) (USEPA, 1989):

$$I_d = \frac{C_w \times SSA \times k_p \times CF \times ET \times EF \times ED}{BW \times AT} \quad \text{Eq. (2)}$$

where SSA is the skin surface area (cm^2); k_p is the permeability coefficient specific for each metal (cm.h^{-1}); CF

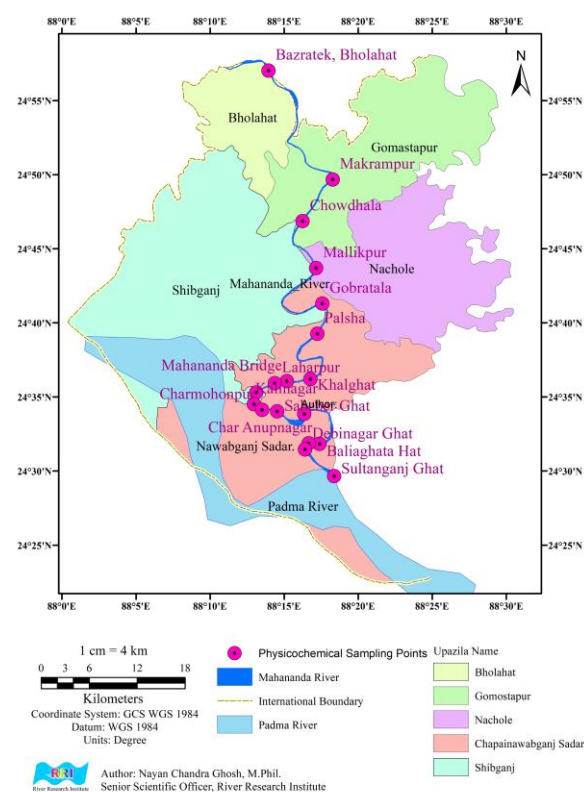


Fig. 1. Surface water sampling points for the analysis of physicochemical parameters.

Table 1. Coordinates of the surface water sampling points for the analysis of physicochemical parameters.

Name	Latitude	Longitude
Bazratek, Volahat	24°57.025'N	88°13.953'E
Makrampur	24°49.673'N	88°18.295'E
Chowdhala	24°46.869'N	88°16.257'E
Mallikpur	24°43.700'N	88°17.171'E
Gobratola	24°41.304'N	88°17.593'E

Name	Latitude	Longitude
Palsha	24°39.265'N	88°17.247'E

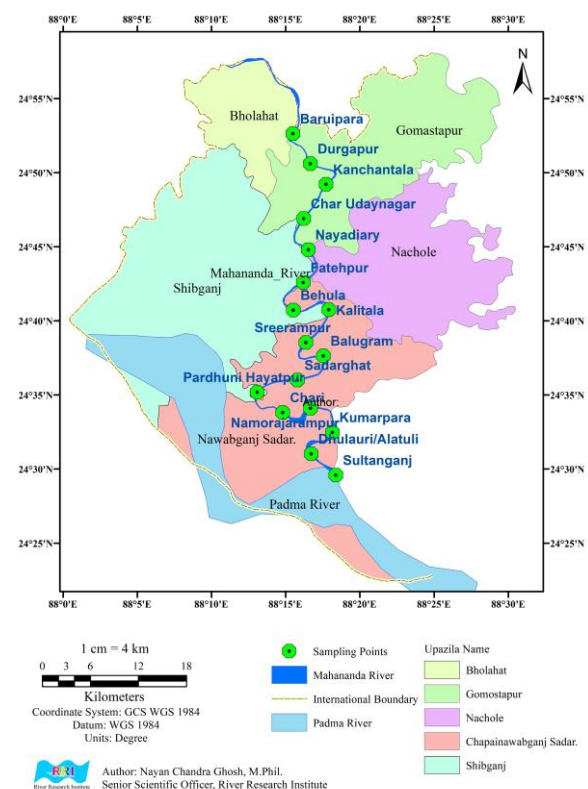


Fig. 2. Surface water sampling points for the analysis of nutrients and elements.

Table 2. Coordinates of the surface water sampling points for the analysis of nutrients and elements.

Name	Latitude	Longitude
Baruipara	24°52.639'N	88°15.490'E
Durgapur	24°50.615'N	88°16.673'E
Kanchantala	24°49.218'N	88°17.724'E
Char Udaynagar	24°46.906'N	88°16.219'E
Nayadiary	24°44.809'N	88°16.532'E
Fatehpur	24°42.595'N	88°16.192'E
Behula	24°40.735'N	88°15.515'E
Kalitala	24°40.760'N	88°17.911'E
Sreerampur	24°38.547'N	88°16.366'E
Balugram	24°37.642'N	88°17.547'E
Sadarghat	24°36.008'N	88°15.792'E
Pardhuni Hayatpur	24°35.191'N	88°13.076'E
Chari	24°33.817'N	88°14.809'E
Namorajampur	24°34.105'N	88°16.680'E
Kumarpara	24°32.479'N	88°18.151'E
Dhulauri/Alatuli	24°31.032'N	88°16.721'E

Name	Latitude	Longitude
Sultanganj	24°29.598'N	88°18.384'E

Table 3. Risk assessment parameters.

Parameters	Value (Adult)	Value (Child)	
<i>IR</i>	2	1	(Akter et al., 2021)
<i>EF</i>	365	365	(USEPA, 1989)
<i>ED</i>	74.3	10	(BBS, 2014; Akter et al., 2021)
<i>BW</i>	70	25	(Akter et al., 2021)
<i>AT</i>	27120	3650	(USEPA, 2011)
<i>SSA</i>	18000	13000	(USEPA, 2011)
<i>k_p</i>	0.001	0.001	(USEPA, 2004)
<i>CF</i>	0.001	0.001	(Turdiyeva and Lee, 2023)
<i>ET</i>	0.58	1.00	(USEPA, 2004)
<i>SF</i> oral	1.5	1.5	(USEPA IRIS, 2019; RAIS, 2022; OEHA, 2018)
<i>SF</i> dermal	1.58	1.58	(USEPA, 2004)
<i>RfD</i> oral (As)	0.0003	0.0003	(USEPA IRIS, 2019; RAIS, 2022)
<i>RfD</i> dermal (As)	0.000285	0.000285	(USEPA, 2004)
<i>RfD</i> oral (Mn)	0.024	0.024	(USEPA, 2004; USEPA IRIS, 2019; RAIS, 2022)
<i>RfD</i> dermal (Mn)	0.00096	0.00096	(USEPA, 2004)

Results and Discussions

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Conclusions

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Recommendations

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Acknowledgment

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Conflict of Interest

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Author's Contribution

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TABLE ANATOMY

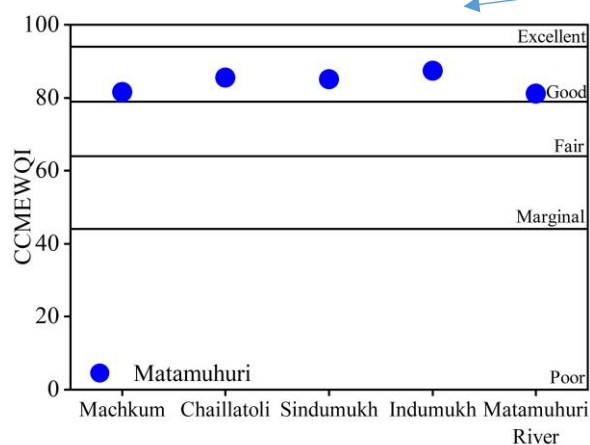
Table 4. Summary of the test scenarios in the fixed bed model*Table caption and Table body (9 font)*

Test	Objectives	Scenarios	Flow Condition
T0-1	Trial calibration of the model	September 2016 bathymetry with existing structures (Jetty)	Q=11489 m ³ /s(ebb) Q=6600m ³ /s(tide)
T0-2	Calibration of the model	September 2016 bathymetry with existing structures (Jetty)	Q=11489m ³ /s(ebb) Q=6600m ³ /s(tide)
T1	To test the performance of the proposed sheet pile and revetment along with Paira bridge and consequent hydraulic and morphological changes.	September 2016 bathymetry with existing structure (Jetty) in position plus proposed sheet pile + revetment and Paira bridge	Q=13861m ³ /s(50- year design discharge) Q=6600m ³ /s(tide)
T2	To test the performance of revised design of revetment along with Paira bridge and consequent hydraulic and morphological change.	September 2016 bathymetry with existing structures (Jetty) plus proposed revetment and Paira bridge	Q=13861m ³ /s(50- year design discharge) Q=6600m ³ /s(tide)
T3-a	To test the performance of final design of revetment and dredged channel along with Paira bridge and consequent hydraulic and morphological changes around the bridge corridor	September 2016 bathymetry with existing structures (Jetty) plus revetment and Paira bridge. 900m long proposed dredged channel is introduced along the right side of Pier- 18.	Q=12698m ³ /s (50- year design discharge as a reverse flow), WL=3.24mPWD. Q=6600m ³ /s(tide), WL=0.79 mPWD

Note: Water Levels (WL) are based on the datum of the Public Works Department (PWD). The PWD datum is 0.46 m higher than the mean sea level (Tingsanchali and Karim, 2005)

Table footnote (8 font)

GRAPH



Please, don't write any graph heading/ caption over here. It looks very ugly

Fig. 5. CCMEWQI of Matamuhuri Rivers.*Figure Caption (9 font)*

IMPORTANT NOTES:

Please **avoid** sentences that **give no information** other than directing the reader to the Figure or Table.

For instance,

Scale factors for the prototype and model have been obtained for different parameters are presented in the following

Table 1

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On the basis of the above considerations the scale factors for the prototype and model have been obtained for different parameters (**Table 1**).

SIMILAR FIGURES

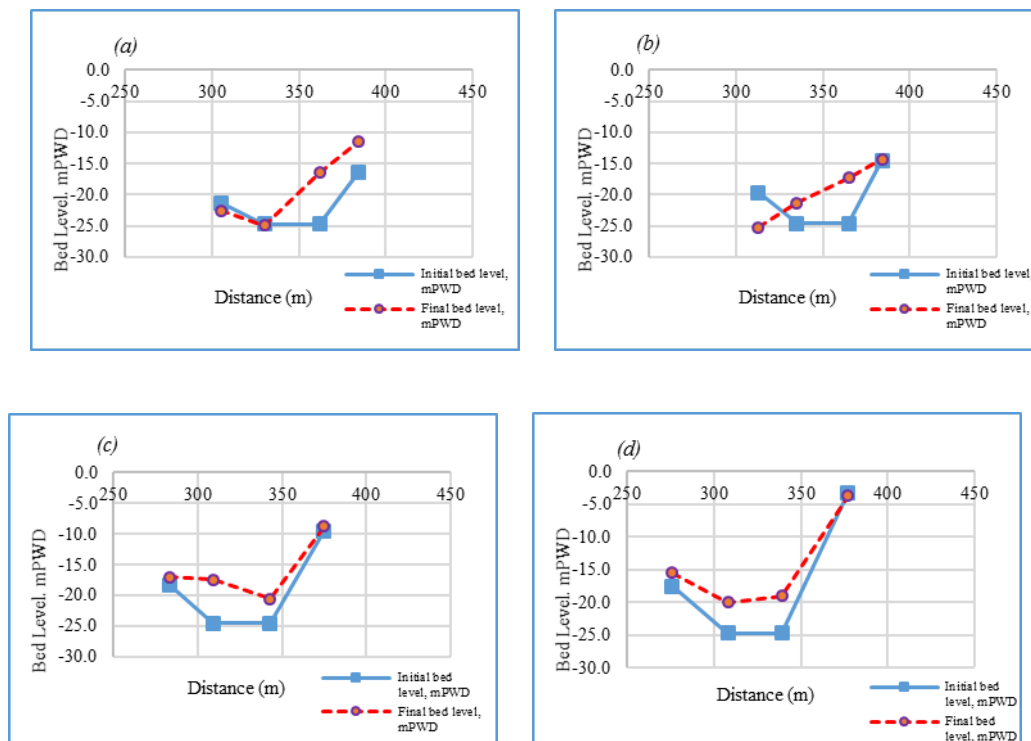


Fig. 2. The initial and final bed level of dredged channel (a) along cross-section 10 US (b) along cross-section 08US (c) along cross-section 06 US (d) along cross-section 04 US

Note: Similar figure should be clustered under a common caption like above, but every figure should be clearly specified

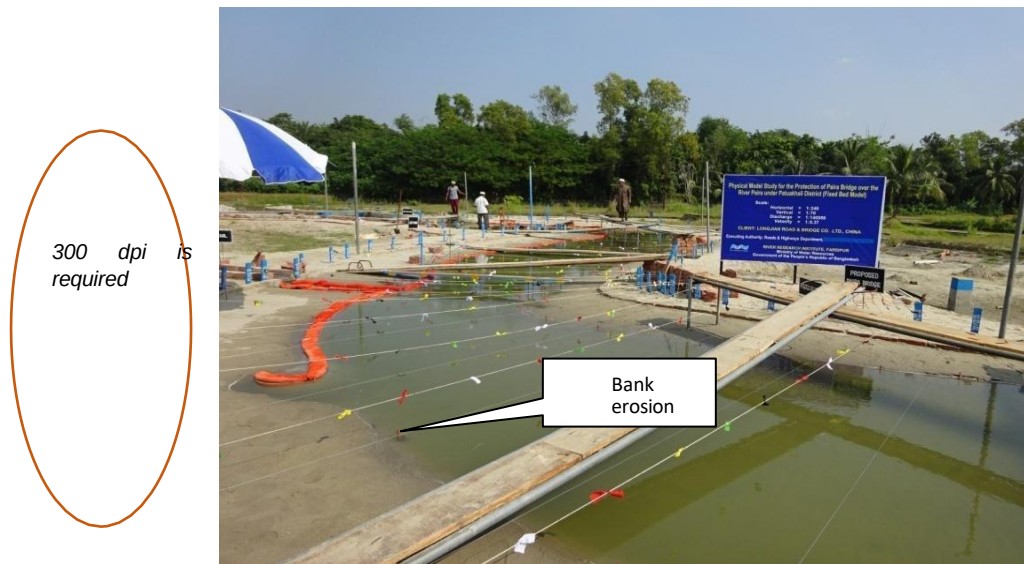


Fig. 3. Bank erosion tendency at the upstream of the proposed revetment for test T1

Note: Dot per inch. Dpi (actually, pixel per inch. ppi) is important in case of printing.

UNITS, ABBREVIATIONS, SYMBOLS:

Only metric units (**SI**) should be used in a manuscript. After the first appearance of a term in full, a standard abbreviation may be used. Please take care that all terminology and notation used will be widely understood. Superscripts, **not slashes (/)**, should be used to describe units, e.g. **m³s⁻¹**, **kg m⁻³ s⁻¹**, **m s⁻¹**.



EQUATIONS:

Mathematical expression/equation should be inserted using Microsoft Word. Equations should be numbered consecutively and referenced in the text [e.g. **Eq. (1)**],

for example,

$$Q = (0.403 + 0.53h_w/p)(2g)^{0.5}b [(h_w + u^2/2g)^{1.5} - (u^2/2g)^{1.5}] \quad \text{Eq. (1) (not bold)}$$

Equation numbering style (not bold)

A SECTION WITH SEVERAL POINTS

Each test of the model continues about 16-20 hours until a dynamic equilibrium scour is reached. Considering the above issues the model scale is selected ensuring the following model requirements.

- (a) The model scale is undistorted.
- (b) The model sale is selected to ensure sufficient water depth for measurement of velocity.
- (c) The model fulfilled flow as well as sediment transport condition simultaneously.
- (d) The geometric scale should be such that it allows the model to be fitted within the available laboratory space at RRI.
- (e) The discharge requirement in the model should be within RRI pumping capacity.

Numbering style (don't insert any bullet point like
or any other style) •

RESULTS AND DISCUSSIONS

Results should be clear and abridged. Discussions should explore the consequences of the outcomes of the work, don't repeat them again. Authors can write the Results and Discussions section collectively.

CONCLUSION

The main conclusion of the research should be clear as well as concise. Conclusion may be presented in a short conclusion section.

ACKNOWLEDGEMENTS

Please mention those individuals (if any) who provided help either financial or any other form during research. This section should be placed in a distinct segment towards the end of the article afore the references.

Conflict of Interest

Please mention whether there is any conflict of interest among the authors or other persons.

Author's Contributions

Contribution of each authors in the manuscript must be specified clearly.

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Citing in the text References should be cited in the text either in brackets, e.g. Earlier studies (Mehedi, 2014) showed... or using the name **as part of a sentence**, e.g. Mehedi (2014) states...

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In reference list

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Journal articles

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